



Major Diseases of Rice and their management

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Abstract

Rice (*Oryza sativa* L.) is one of the most important staple food crops of the world nearly half of the world's population depends on rice. Rice plays an important role in global food security, it is prone to attack by many diseases caused by plant pathogens such as fungi, bacteria viruses, and nematodes. Among diseases, rice blast, bacterial leaf blight, brown leaf spot, sheath blight, sheath rot, stem rot, false smut, rice tungro virus, and rice root-knot nematode are economically important and they cause huge economic losses. Recently, due to changes in the global temperature and change in climatic situation minor diseases are becoming major (false smut of rice and rice root-knot nematode), The various methods used for managing rice disease includes, the use of resistant varieties, cultural practices, biological and chemical control.

All these methods have varied degrees of success in managing rice diseases. The most important management tactics used worldwide include the use of resistant varieties and chemical control. Integration of all the management tactics (Integrated Disease Management) could help in the effective and successful management of plant disease in rice. In this chapter, we are showcasing the important diseases of rice, global distribution, economic importance and management strategies.

Keywords: Rice, Integrated Disease Management, Disease, Viruses

Introduction

Rice (*Oryza sativa* L.) is the world's most significant cereal crop and also the fastener food of over half of the world's people. Rice rank second among the most cereals in terms of total grain production after maize worldwide.

Rice is cultivated on about 169 million hectares with 780.9 million tonnes of total world production. India is a leading country and donate in area and production of rice is 43.79 mh and 116.42 mt, respectively with average productivity of 2659 kg/ha. Among the chief yield-limiting factors pests are said to be a significant one. Pest causes 33% of production losses in India, the most important pest weed causes 12.5 % whereas insect 9.5% and disease 6.5%.

There are several constraints in the production of rice crops. It is affected by different biotic and abiotic factors which are answerable for its low yield. Among the biotic factors disease caused by fungi, bacteria, viruses, nematodes and insects are responsible for major economic losses in diverse parts of our country.

In this view, IDM is a science-based approach to managing pest and diseases by considering the environment, the pest or disease, and the conditions that encourage survival, the crop and its growing situation and nature's ecology and relations with natural enemies. The use of biodegradable pesticides is only applied and used as a last remedy based on careful environmental monitoring. Here we will talk about some significant diseases of rice and their management in an integrated way.

1. Brown spot

Brown spot is the most imperative fungal disease of rice that can infect both seedlings and full-grown plants. The disease causes blight on seedlings, which are grown from greatly infected seeds and can cause 10-58% seedling mortality. The disease was considered to be the major thing contributing to the "*Great Bengal Famine*" in 1942.

Symptoms

- ✓ This disease might be manifested as seedling blight or as a foliar and glume disease of full-grown plants. In seedlings, the fungus produces small, circular brown lesions which may girdle the coleoptiles and cause alteration of the leaves. Infected seedlings are undersized or killed.
- ✓ In grown-up plants, the lesions on the leaves are light brown to gray in the center and have a reddish brown margin and lesions may vary between 1mm to 14mm long. Later, lesions may unite in harsh infections killing large areas of pretentious leaves.
- ✓ The fungus be capable of infecting the glumes and the grains, thus reducing the number of grains per panicle and the kernel weight.



Fig: Brown spot of rice

Causal organism

The causal organism of brown spot of rice is *Bipolaris oryzae*, *Cochilobolus miyabeans* (synonyms *Dreschlera oryzae*, *Helminthosporium oryzae*). The somatic structures of the fungus consist of black velvety mycelial mats which are made up of desperate hyphae and erect sporophores. The hyphae are abundant, branching, and anastomosing. They are dark brown or olivaceous and measure 8-15 cm or more in diameter.

Disease cycle and Epidemiology

Impure seeds are the most ordinary source of primary infection. The fungus also survives on collateral hosts. It can stay alive in the seed for over 4 years. The fungus is able to extend from plant to plant and in the field by airborne spores. The most favorable temperature ranges from 25°C -30°C with high humidity 86-100%.

Management

- ✓ Give well-balanced nutrients to the soil.
- ✓ Clean all rice garbage and weeds from fields which sources of inoculum.
- ✓ Avoid water stress conditions and use resistant/tolerant cultivars
- ✓ The fungus be capable of surviving on seeds for up to 4 years, therefore, seed treatment with captan.
- ✓ Soaking seed in hot water 53-54°C for 10-12 minutes may also be effective.
- ✓ Seed treatment with tricyclazole followed by spraying of mancozeb + tricyclazole at tillering and late booting stages give good management of the disease.

2. Rice blast

Rice blast is a significant disease worldwide and known since most primitive times as an overwhelming problem in Asia. The disease is also called as rotten neck or rice fever in China. In India, its first incidence was recorded in 1918. The disease is predicted to cause grain loss up to 70 to 80%.

Symptoms

Blast is a foliar disease in violent above-ground plant parts but the majority is found on leaves (leaf and collar blast) or the panicles (neck and panicle blast). On top of the leaves, very untimely lesions (spots) are off-white to gray-green in color with a dark green border and only 1/16 - 1/8" across. These spots quickly twist gray or gray-white in the center with a brown-to-reddish-brown border.

As regards a week after infection, the fungus sporulates inside the lesion producing a dark, gray powdery appearance, especially in near the beginning of the morning or on the underside of the lesion during the day. It is very general for blast to attack the bottom of the flag leaf (collar blast), prematurely killing it, and infecting the sub-panicle node as it exerts through this infected area. At caption, blast attacks the node just below the panicle (neck blast) killing the entire panicle before grain fill is complete.



Fig: Rice blast

Causal pathogen

Blast is caused by the fungus, *Pyricularia grisea* (formerly *Pyricularia oryzae*). The fungus has a sexual stage known as *Magnaporthe grisea*, which has simply been observed in the laboratory.

Disease cycle and Epidemiology

The rice blast fungus only attacks rice and has no other plant host. Intimately-related forms of the blast fungus infect uncultivated grasses and some turf grasses but do not infect rice.

The rice blast fungus survives among rice crops in infested rice residue or on seed. Because the spores are airborne, the faction between fields through the season is familiar and rapid. Cooler temperatures than normal, cloudy, and rainy weather enhance blast incidence and severity in susceptible or moderately vulnerable cultivars.

Management

- ✓ Use resistant varieties such as CO 47 and remove collateral weed hosts from bunds and channels.
- ✓ Use only disease-free seedlings and avoid excess nitrogen.
- ✓ Apply N in three split doses (50% basal, 25% in tillering phase and 25% N in panicle initiation stage).
- ✓ Spray after observing initial infection of the disease, Carbendazim 50WP @ 500g/ha (or) Tricyclozole 75 WP @ 500g/ha.

3. Rice sheath blight

Sheath blight of rice is one of the major and devastating diseases of rice subsequently to blast. The disease was earliest reported in Japan by Miyake in 1910. Later, its incidence was recorded all over the temperate and tropical rice-growing areas, worldwide. In India, the disease was earliest reported by Paracerand Chahalin 1963 from Gurdaspur, Punjab. The disease causes as regards 11.1-58% losses in yield depending on the disease brutality and varietal resistance.

Symptoms

Primarily the disease shows as elliptical or oval to irregular 1-3 cm long, greenish gray spots with brown margin at or on top of the water line on leaf sheath at active tillering stage which later gives the look of snakeskin. The infected plants are typically found in a circular blueprint, locally referred to as a 'bird's nest'.

Under damp conditions, brown silky mycelium and brown to dark brown sclerotia are originate.

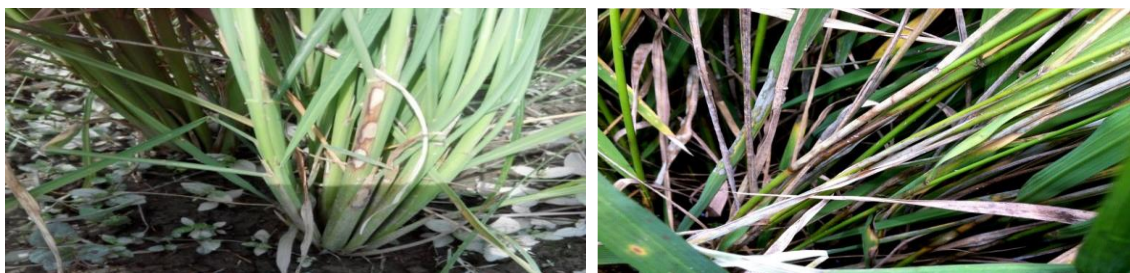


Fig: Rice sheath blight

Causal pathogen

Sheath blight is a fungal disease of rice caused by a necrotrophic soil-borne fungus *Rhizoctonia solani* with telomorphic stage *Thanatephorus cucumeris* freely attached on the lesions, which obtain simply dislodged from the plants at adulthood. The immature hyphae are colourless but happen to yellow and eventually brown with age, 8-12 cm. The figure of the sclerotia is roughly spherical or rather flattened and unequal. Young sclerotia are possess of solid masses of hyphal cells about 5 cm wide, the cell wall 0.9 cm thick.

Disease cycle and Epidemiology

Sheath blight is a customized single cycle disease. The pathogen survives in soil inside crop garbage as mycelia or sclerotia during unfavourable ecological conditions for numerous years. Subsequent to rice harvest, *R. solani* sclerotia from infected plants endure in the soil for periods of up to 3 years and act as a basis of infection for succeeding crop cycles.

Highest air temperature, morning relative humidity and leaf dampness have been reported to be the key factors for disease expansion in the field. The highest disease development was recorded at a temperature range of 25-30°C and 80-100 % relative humidity.

Management

- ✓ Summer deep ploughing and burning of stubbles.
- ✓ Utilize a reasonable level of fertilizer and destruction of weeds.
- ✓ Breeding disease-resistant rice cultivars is believed to be one of the most hopeful approaches to manage the disease.
- ✓ Soil amendment with organic manures like *Sesbania aculate*, and *Gliricidia aculate* leaves were efficient against sheath blight.
- ✓ Seed treatment by Bavistin and Benlate, soil appliance of Benlate, Kitazin and Hinosan, foliar use of Topsin-M has been reported successful in reducing this disease.

4. False smut

The false smut of rice is a significant disease that was formerly recorded as a negligible disease of rice but at the present, it occurs in a further 40 countries, particularly in the rice-producing countries of Asia, although also in the U.S.

Symptoms

As the familiar name suggests, it is not a true smut (fungus) as it belongs to Ascomycotina. False smut does not swap all or components of the kernel with a mass of black spores. Plants tainted with false smut have individual rice grains, altered into a mass of spore balls. Such spore balls are primarily 1 cm in size and orange in color, which revolve into greenish black when these grown-ups.



Fig: False smut disease of rice

Causal organism

The Rice False Smut (RFS) is caused by *Ustilagi noidea virens*. The conidia are oblique with diameters ranging from 3 to 5 μ m. Ahead of maturation or under adverse conditions, conidia may build up to rounded chlamydospores with well-known spines on the shell.

Disease cycle and Epidemiology

U. virens has a strange life cycle. The fungus attacks rice flowers and forms balls enclosed with chlamydospores and produces sclerotia, which are considered as crucial inoculum of this disease. During its life cycle, *U. virens* undergoes a sexual (ascospores) stage as well as an asexual (chlamydospores) stage. They take shelter in the field and might survive up to 10 months with maintaining germination capability to produce ascospores under 25°C and elevated humidity.

Management

- ✓ Use of disease-free seeds that are selected from healthy crops.
- ✓ Split relevance of nitrogenous fertilizer.
- ✓ Elimination and appropriate disposal of contaminated plant debris.
- ✓ At the time of harvesting, sickly plants should be detached and shattered so that sclerotia do not fall in the field.
- ✓ Eliminate alternate weed hosts.
- ✓ Seed treatment by carbendazim 2 g/kg of seeds.
- ✓ Spraying of copper oxychloride at 2.5 g/litre or Propiconazole at 1.0 ml/litre at boot leaf and milky stages will be more useful to put off the fungal infection.

5. Bacterial Leaf Blight (BLB)

Rice bacterial blight, also well-known as the bacterial blight of rice. It is amongst the cruelest disease of cultivated rice (*Oryza sativa* and *O. glaberrima*). Bacterial blight has been realistic in rice-emergent regions, prosperous in warm and damp environments such as Asia, the western coast of Africa, Australia, Latin America, and the Caribbean. In malicious epidemics, crop hammering may be as eminent as 75 percent, and millions of hectares of rice are infected annually.

Symptoms

The typical symptoms of this disease come into view in two stages: seedling blight or kresek phase and Leaf blight phase.

Seedling wilt or kresek Phase

This phase might be pragmatic at 1-3 weeks subsequent to transplanting. Firstly, green water-soaked film along the cut section or leaf tip leaves are characterized as premature symptoms. Afterward, leaves wilt and roll up and turn into the grayish green to yellow, ensuing absolute wilting of the whole plant.

Leaf blight Phase

The symptoms initially become visible as water-soaked to yellowish stripes on leaf blades or starting on leaf tips by a curvy margin. Such tainted leaves with undulated yellowish white or golden yellow trivial necrosis, drying flipside as of tip and curling, leaving mid rib intact are the most important symptoms. Near the beginning of morning, a bacterial ooze appeared that resembled a milky or dense droplet on infantile lesions. Rigorously infected leaves be predisposed to dehydrated quickly.



Fig:- Bacterial leaf blight of rice

Disease cycle and Epidemiology

Xanthomonas oryzae survives throughout rice kernel, impure crop debris to facilitate are gone behind at harvest, in addition to unconventional weed hosts. *X. oryzae* lives on lifeless plants and seeds are most likely moves plant-to-plant best throughout patty water commencing

irrigation or storms. The pathogen grew healthy on temperature ranges 26-30°C; 20°C being the finest temperature for early growth.

Management

- ✓ Grow Tolerant varieties (IR 2, IR 72, TKM 6).
- ✓ Avoid clipping of seedlings during transplanting.
- ✓ Unbiased fertilization, avoid excess N – application.
- ✓ Destruction of alternate weeds and collateral hosts and maintain proper plant spacing.
- ✓ Seed treatment - seed soaking for 8 hours in Agrimycin (0.025%) and wettable ceresan (0.05%) followed by hot water treatment for 30 min at 52-54°C.
- ✓ Spray Streptomycin sulphate + Tetracycline combination 300 g + Copper oxychloride 1.25kg/ha. If necessary, repeat 15 days later.

6. Bakanae /foolish disease

The most primitive recognized statement of *bakanae* is from 1828; it was earliest described systematically in 1898 by Japanese researcher Shotaro Hori. In outbreak cases yield wounded could reach up to 20% or more.

Symptoms

The impure plants are several inches taller than normal plants, etiolated and chlorotic. They are too thin, by means of yellowish green and pale green foliage. Premature infection be able to cause seedlings to pass away at early tillering stage. Afterward infection fall out in plants that build up only some tillers and include dry leaves. If the plants live on to adulthood stage, they grow partially filled grains, sterile or vacant grains.



Fig :- Foolish disease of rice.

Causal organism

The causal organism was earliest identified by Hori (1808) as *Fusarium hetrosporium*. The sexual stage of fungus was described as *Lisea fugikuroi* by Sawada (1917) which was passed in 1931 to *Gibberella fusikuroi* by Ito and Kimura, who besides recognized the asexual phase as *Fusarium moniliforme*.

Disease cycle and Epidemiology

It is a monocyclic disease and the pathogen is both soil and seed-borne. The most advantageous warmth for pathogen infection is 27- 30°C and for disease, is further considerable source as soil-borne inoculum is reduced rapidly passage of time.

Management

- ✓ Make use of salt water to separate lightweight, infected seeds during soaking.
- ✓ Exploit fungicides as seed treatments. applying a fungicide containing benomyl or benomyl-t (at 1-2% of seed weight) for dry seed covering to treat contaminated seeds can be efficient.
- ✓ Soaked seed in a fungicide solution of 1:1000 for one hour or 1:2000 for five hours has also been shown to be useful.

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